

Correlation Questions

Correlation played a central role in the accuracy calculations of Chapter 7, with the estimated root mean square correlation $\hat{\alpha}$ being the key quantity in approximations (7.2), (7.49), and (7.50). This chapter takes up several questions concerning correlation in large-scale studies, including the calculation of $\hat{\alpha}$, which often turns out to be quite straightforward. The setting will be like that for our microarray examples: we observe an $N \times n$ matrix X as in (2.1); the columns of X correspond to individual subjects in Section 2.1, hopefully sampled independently of each other, while we expect the gene expressions within a column to be correlated.

Independence of the columns of X is a tacit assumption of all theoretical and permutation null hypothesis testing methods (calculations like (2.2)–(2.6) make no sense otherwise), though not of empirical null procedures. Section 8.3 considers tests for column-wise independence. We begin in Section 8.1 with a general result relating the row-wise and column-wise correlations of X , leading directly to the estimator $\hat{\alpha}$.

8.1 Row and Column Correlations

We will be interested in both the row-wise and column-wise correlations of the $N \times n$ matrix¹ $X = (x_{ij})$. It simplifies notation to assume that X has been “demeaned” by the subtraction of row and column means, so that

$$\sum_{i=1}^N x_{ij} = \sum_{j=1}^n x_{ij} = 0 \quad \text{for } i = 1, 2, \dots, N \text{ and } j = 1, 2, \dots, n. \quad (8.1)$$

¹ The number of rows of X is often denoted by p rather than N in the microarray literature, following (in transposed fashion) an analogy with the familiar ANOVA $n \times p$ structure matrix. This can cause confusion with p -values and probabilities. Our use of “big N ” suggests the usual microarray situation $N \gg n$.

Sometimes we will go further and assume *double standardization*: that in addition to (8.1) we have

$$\sum_{i=1}^N x_{ij}^2 = N \quad \text{and} \quad \sum_{j=1}^n x_{ij}^2 = n \quad (8.2)$$

for $i = 1, 2, \dots, N$ and $j = 1, 2, \dots, n$.

Double standardization is convenient for our discussion because it makes sample covariances into sample correlations. How to achieve (8.2) is discussed at the end of this section.

With means removed as in (8.1), the sample covariance between columns j and j' of X is

$$\hat{\Delta}_{jj'} = \sum_i x_{ij} x_{ij'} / N \quad (8.3)$$

(not reducing the denominator by 1) and similarly

$$\hat{\sigma}_{i'j} = \sum_j x_{ij} x_{i'j} / n \quad (8.4)$$

for the sample covariance between rows i and i' . The $n \times n$ column sample covariance matrix $\hat{\Delta}$ and the $N \times N$ row sample covariance matrix $\hat{\Sigma}$ are

$$\hat{\Delta} = X'X/N \quad \text{and} \quad \hat{\Sigma} = XX'/n. \quad (8.5)$$

Double standardization makes the diagonal elements $\hat{\Delta}_{jj}$ and $\hat{\sigma}_{ii}$ equal 1. Then the sample correlation $\hat{\rho}_{i'j}$ between rows i and i' of X is

$$\hat{\rho}_{i'j} = \hat{\sigma}_{i'j} / (\hat{\sigma}_{ii} \hat{\sigma}_{j'j})^{1/2} = \hat{\sigma}_{i'j}. \quad (8.6)$$

Our main result is stated and proved in terms of the *singular value decomposition* (svd) of X ,

$$X = \underset{N \times n}{U} \underset{N \times K}{\mathbf{d}} \underset{K \times K}{V'} \underset{K \times n}{V'} \quad (8.7)$$

where K is the rank of X , $\mathbf{d}$ is the diagonal matrix of ordered *singular values* $d_1 \geq d_2 \geq \dots \geq d_K > 0$, and U and V are orthonormal matrices of sizes $N \times K$ and $n \times K$,

$$U'U = V'V = I_K, \quad (8.8)$$

with I_K the $K \times K$ identity. The squares of the singular values,

$$e_1 \geq e_2 \geq \dots \geq e_K > 0 \quad [e_k = d_k^2] \quad (8.9)$$

are the eigenvalues of $N\hat{\Delta} = V\mathbf{d}^2V'$ and also of $n\hat{\Sigma} = U\mathbf{d}^2U'$.

The singular value decomposition reveals an interesting correspondence between the row and column correlations of X .

Theorem 8.1 If X has row and column means equal to zero (8.1), then the n^2 elements of $\hat{\Delta}$ and the N^2 elements of $\hat{\Sigma}$ both have mean zero and empirical variance

$$c_2 = \sum_{k=1}^K e_k^2 / (Nn)^2. \quad (8.10)$$

Proof Letting $\mathbf{1}_n$ denote the vector of n 1's, the sum of the entries of $\hat{\Delta}$ is

$$\mathbf{1}_n' X' X \mathbf{1}_n / N = 0 \quad (8.11)$$

according to (8.1), while the mean of the squared entries is

$$\frac{\sum_{j=1}^n \sum_{j'=1}^n \hat{\Delta}_{jj'}^2}{n^2} = \frac{\text{tr}((X'X)^2)}{(Nn)^2} = \frac{\text{tr}(V \mathbf{d}^4 V')}{(Nn)^2} = c_2. \quad (8.12)$$

Replacing $X'X$ with XX' yields the same result for $\hat{\Sigma}$. $\square$

Exercise 8.1 Here tr indicates the trace of a square matrix, the sum of its diagonal elements. Verify (8.12). *Hint:* Use the trace commutative property $\text{tr}(AB) = \text{tr}(BA)$.

The covariances $\hat{\Delta}_{jj'}$ and $\hat{\sigma}_{ii'}$ become correlations under double standardization, as in (8.6). Then Theorem 8.1 says that the row and column sample correlations have the same means and empirical standard deviations. This sounds definitely counter-intuitive in a typical microarray setting, where the genes (rows) are assumed correlated and arrays (columns) independent.

Figure 8.1 shows an example. Here the data is from the *Cardio Study*, a microarray experiment comparing $n = 63$ subjects, $n_1 = 44$ healthy controls and $n_2 = 19$ cardiovascular patients, each measured on $N = 20\,426$ genes. The matrix X used in Figure 8.1 is the doubly standardized expression data for the healthy controls, so X is $20\,426 \times 44$ and satisfies (8.1)–(8.2).

The solid histogram shows all 44^2 column sample correlations, while the line histogram is a random sample of 10 000 row correlations (out of the more than 400 million values $\hat{\rho}_{ii'}$!). In this case c_2 (8.8) equals 0.283^2 , so both histograms have mean zero and standard deviation 0.283.

Exercise 8.2 What is a computationally efficient way to calculate c_2 (without using the SVD)? *Hint:* The trace of a square symmetric matrix is the sum of its eigenvalues.

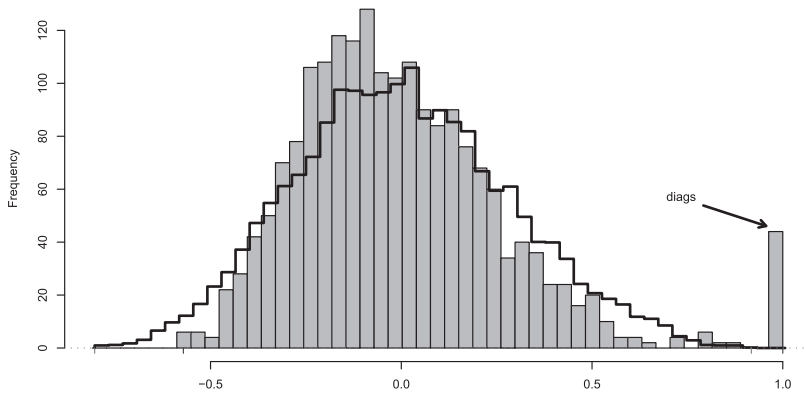


Figure 8.1 Solid histogram shows the 44^2 column sample correlations of the doubly standardized expression matrix X for the healthy cardio controls. Line histogram depicts the sample of 10 000 of the $20\,426^2$ row correlations. Both histograms have mean 0 and standard deviation 0.283.

The 44 diagonal elements of $\hat{\Delta}$, the column correlation matrix, protrudes as a prominent spike at 1. (We cannot see the spike of 20 426 elements of $\hat{\Sigma}$, the row correlation matrix, because they form such a small fraction of all $20\,426^2$.) It is easy to remove the diagonal 1's from consideration.

Corollary 8.2 *In the doubly standardized situation, the off-diagonal elements of the column correlation matrix $\hat{\Delta}$ have empirical mean and variance*

$$\hat{\mu} = -\frac{1}{n-1} \quad \text{and} \quad \hat{\sigma}^2 = \frac{n}{n-1} \left(c_2 - \frac{1}{n-1} \right) \quad (8.13)$$

with c_2 as in (8.10).

Exercise 8.3 (a) Verify (8.13). (b) Show that $c_2 \geq 1/K$. *Hint:* Under double standardization, $\bar{e}$, the average of the eigenvalues (8.9), equals Nn/K .

Standardization

A matrix X is *column standardized* by individually subtracting the mean and dividing by the standard deviation of each column, and similarly for row standardization. Table 8.1 shows the effect of successive row and column standardizations on the original $20\,426 \times 44$ matrix of healthy cardio

subjects. Here “Col” is the empirical standard deviation of the 946 column-wise correlations $\hat{\Delta}_{jj'}$, $j < j'$; “Eig” is $\hat{\alpha}$ in (8.13); and “Row” is the empirical standard deviation of a 1% sample of the row correlations $\hat{\rho}_{ii'}$, adjusted for overdispersion as in (8.18) of the next section. The sampling error of the Row entries is about ± 0.0034 .

Table 8.1 *Estimates of the rms correlation $\hat{\alpha}$ after successive column and row standardization of the $20\,426 \times 44$ matrix of healthy cardio subjects, as explained in the text. Sampling error of Row entries ± 0.0034 .*

	Col	Row	Eig		Col	Row	Eig
demeaned	.252	.286	.000	demeaned	.252	.286	.000
col	.252	.249	.251	row	.241	.283	.279
row	.242	.255	.246	col	.241	.251	.240
col	.242	.241	.242	row	.240	.247	.241
row	.241	.246	.235	col	.240	.247	.240
col	.241	.244	.241	row	.241	.240	.235
row	.241	.245	.234	col	.241	.237	.240
col	.241	.238	.241	row	.241	.233	.233

The doubly standardized matrix X used in Figure 8.1 was obtained after five successive column/row standardizations of the original cardio matrix. This was excessive: the figure looked almost the same after two iterations. Other microarray examples converged equally rapidly, and it can be shown that convergence is guaranteed except in some special small-size counterexamples.

Microarray analyses usually begin with some form of column-wise standardization, designed to negate “brightness” differences among the n arrays. In the same spirit, row standardization helps prevent incidental gene differences (such as very large or very small expression level variabilities) from obscuring the effects of interest. This isn’t necessary for z -values based on t statistics, which are “self-standardizing,” but can be important in other contexts, as demonstrated in Chapter 10.

8.2 Estimating the Root Mean Square Correlation

For $n = 44$ and $c_2 = 0.283^2$, formula (8.13) yields

$$(\hat{\mu}, \hat{\alpha}^2) = (-0.023, 0.241^2), \quad (8.14)$$

reducing our estimated standard deviation for the column correlations from 0.283 to 0.241. The corresponding diagonal-removing correction for the row correlations (replacing n by N in (8.13)) is negligible for $N = 20426$. However, c_2 overestimates the variance of the row correlations for another reason: with only $n = 44$ bivariate points available to estimate each correlation, random error adds a considerable component of variance to the $\hat{\rho}_{ii'}$ histogram in Figure 8.1. Once this overdispersion is corrected for, $\hat{\alpha}$ in (8.13) turns out to be a reasonable estimator for α , the root mean square correlation (7.39).

We assume that the columns $\mathbf{x}$ of $X = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$ are independent and identically distributed N -vectors, with correlation $\rho_{ii'}$ between entries x_i and $x_{i'}$. The rms correlation is

$$\alpha = \left[\sum_{i < i'} \rho_{ii'}^2 / \binom{N}{2} \right]^{\frac{1}{2}} \quad (8.15)$$

which is another notation for (7.39).

Define $\bar{\alpha}$ to be the observed row-wise root mean square correlation,

$$\bar{\alpha} = \left[\sum_{i < i'} \hat{\rho}_{ii'}^2 / \binom{N}{2} \right]^{\frac{1}{2}}. \quad (8.16)$$

In the doubly standardized case,

$$\bar{\alpha}^2 = \frac{Nc_2 - 1}{N - 1} \doteq c_2 \quad (8.17)$$

but definition (8.16) can be evaluated directly, with or without standardization. (Program alpha, Appendix B, approximates α by randomly selecting a subset I of the rows of X , evaluating $\hat{\rho}_{ii'}$ for i and i' in I , and calculating the corresponding root mean square $\hat{\rho}_{ii'}$ value.) Our preferred estimator for the rms correlation, the one used in Figure 7.1, is

$$\hat{\alpha} = \left[\frac{n}{n-1} \left(\bar{\alpha}^2 - \frac{1}{n-1} \right) \right]^{\frac{1}{2}}. \quad (8.18)$$

Under double standardization this is almost the same as $\hat{\alpha}$ in (8.13). Section 8.4 justifies (8.18) squared as a nearly unbiased estimator of the true α^2 in multivariate normal models, a result supported more generally by numerical simulations. It gave $\hat{\alpha} = 0.245$ for the cardio example.

Table 8.2 shows $\hat{\alpha}$ estimates for the leukemia data of Figure 6.1a, applied separately to the two patient classes and also to the entire 7128×72 expression matrix. The original matrices were first standardized by having

Table 8.2 $\hat{\alpha}$ (8.18) for the leukemia data of Figure 6.1a.

	AML	ALL	Together
standardized	.120	.104	.117
doubly standardized	.110	.088	.103

each column linearly transformed to mean 0 and variance 1; $\hat{\alpha}$ values for the doubly standardized versions, described below, are seen to be somewhat smaller. The “together” calculation is biased upward since genes that actually express differently in the two patient classes will appear to be more highly correlated, though that effect doesn’t seem crucial here. In general, we should remove any suspected “signal” from the rows of X (such as a regression on a known covariate) before estimating the row-wise correlations.

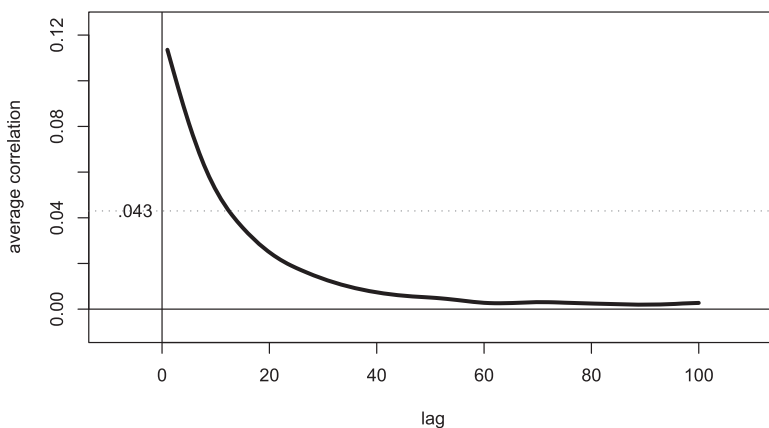


Figure 8.2 Average serial correlations for the snp counts along $N = 26\,874$ sites on chromosome 9 for 313 healthy Caucasian men; rms correlation $\hat{\alpha} = 0.043$.

Figure 8.2 concerns a single nucleotide polymorphism (snp) study that we shall see more of in Chapter 10: $n = 313$ healthy Caucasian men had their genetic bases — A, G, T, or C — read at $N = 26\,874$ sites on both copies of chromosome number 9. The $N \times n$ matrix X had entries x_{ij} equaling the number of bases disagreeing with the modal choice at that site, $x_{ij} = 0, 1$, or 2 , i.e., the number of “snps.” Program alpha gave $\hat{\alpha} = 0.043$,

much smaller than in the cardio or leukemia studies. Figure 8.2 shows the average lagged correlation $\hat{\rho}_{i,i+j}$ for lags $j = 1, 2, \dots, 100$, as we move along chromosome 9. Nearby locations are seen to be more positively correlated. However, most of the rms value $\hat{\alpha} = 0.043$ is not accounted for by serial correlation.

Exercise 8.4 Why do we know this last statement is true?

Table 8.3 *Rms correlation estimates $\hat{\alpha}$ for prostate data, Section 2.1, and HIV data, Section 6.1.*

	Controls	Patients	Together
prostate data	.014	.016	.033
HIV data	.191	.209	.290

Program alpha was used to estimate α for the prostate and HIV microarray studies of Section 2.1 and Section 6.1. Table 8.3 shows $\hat{\alpha}$ to be near zero for the prostate study, and quite large for the HIV data (about as large as $\hat{\alpha} = 0.245$ for the cardio study).

Chapter 7 showed how correlation can drastically degrade estimation accuracy, a point reiterated in Section 8.4. Microarrays are still a relatively new technology, and improved biomedical methods in exposure, registration, and background control may reduce correlation problems. Purely statistical improvements can also reduce correlations, perhaps by more extensive standardization techniques. However, none of this will help if microarray correlations are inherent in the way genes interact at the DNA level, rather than a limitation of current methodology. My own experience has shown microarray studies in general to be the worst correlation offenders; imaging experiments often appear to be highly correlated, as in Figure 2.4, but the correlation is local rather than long-range, yielding small values of the rms correlation.

Simulating correlated z-values

A great advantage of the theory in Chapter 7 is that the rms accuracy formulas depend only on the rms correlation α and not on the entire $N \times N$ covariance matrix Σ for the N -vector of z -values $\mathbf{z}$. However, simulations like that for Figure 7.4 require us to generate vectors $\mathbf{z}$ whose Σ matrix has the desired α . The program `simz` uses a simple construction to do so; the N

cases are divided into J blocks of length $H = N/J$ (with default $J = 5$); independent $N(0, 1)$ variates U_j and V_{hj} are generated for $j = 1, 2, \dots, J$ and $h = 1, 2, \dots, H_j$, and the N z -values are set equal to $(\gamma U_j + V_{hj})/(1 + \gamma^2)^{1/2}$. The constant γ is chosen to make the rms correlation of the demeaned variates $z_i - \bar{z}$ equal the desired value of α .

For the simulation of Figure 7.4, with $N = 6000$ and $\alpha = 0.10$, the correlation matrix of each column of X had five blocks of 1200 “genes” each, correlation 0.20 within blocks and -0.05 across blocks. The 80 columns were generated independently in this way. A constant δ was added to the first 300 genes of the last 40 columns in order to simulate the two-class model (7.42).

Exercise 8.5 X was demeaned for the simulation in Figure 7.4. What was the value of the constant δ ?

8.3 Are a Set of Microarrays Independent of Each Other?

Most of the statistical microarray literature assumes that the columns of X , the individual arrays, are sampled independently of each other, even though the gene expressions within each array may be highly correlated. This was the assumption in Section 2.1, for example, justifying the t -test and the theoretical null hypothesis (2.6). Simple permutation methods (not related to permutation null hypotheses) offer a quick check on the assumption of column-wise independence, sometimes with surprising results.

The scatterplot in Figure 8.3 shows the $N = 20\,426$ points $(x_{i,31}, x_{i,32})$ for the cardio data matrix. It seems to indicate a very strong positive correlation between arrays 31 and 32, with $\hat{\Delta}_{31,32} = 0.805$ (8.3). (Since X is doubly standardized, each row has mean 0, ruling out so-called “ecological correlation,” due only to differing gene-wise response levels.) This seems like a smoking gun argument against column-wise independence, but the gun-smoke isn’t as convincing as it looks, for reasons discussed in Section 8.4.

A simple test for column-wise independence is based on the first eigenvector v_1 of the $n \times n$ column sample covariance matrix $\hat{\Delta}$ (8.5), v_1 also being the first column of V in (8.7). The left panel of Figure 8.4 shows the components of v_1 for the cardio matrix, plotted versus the column number $1, 2, \dots, 44$. Suppose that the columns of the original expression matrix before standardization were independent and identically distributed vectors. This implies that the columns of X are exchangeable, and that all orderings of the components of v_1 are equally likely.

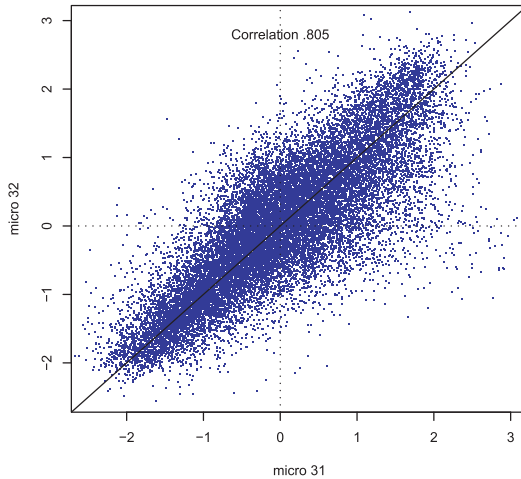


Figure 8.3 Scatterplot of microarrays 31 and 32 of the cardio data, $(x_{i,31}, x_{i,32})$ for $i = 1, 2, \dots, N = 20\,426$. It seems to indicate strong correlation: $\hat{\Delta}_{31,32} = 0.805$.

This is not what the figure shows: the components seem to increase from left to right, with a noticeable block of large values at arrays 27–32. Time trends and block effects are signs of experimental instabilities that cause correlation across arrays. Uncontrolled changes in the development process, for example, can raise or lower expression measurements for whole sets of genes on different areas of the microarray chip, correlating the subsequent block of arrays and raising the corresponding entries of v_1 .

Let $S(v_1)$ be a real-valued statistic that measures structure, for instance the linear regression coefficient of the components of v_1 versus the array index. Comparing $S(v_1)$ with a set of permuted values

$$\{S^{*b} = S(v^{*b}), b = 1, 2, \dots, B\} \quad (8.19)$$

where v^{*b} is a random permutation of the components of v_1 , provides a quick test of the i.i.d. null hypothesis. Taking $S(v_1)$ to be the absolute value of the above-mentioned linear regression statistic, $B = 2000$ permutations yielded only eight values S^{*b} exceeding $S(v_1)$. This gave p -value $8/2000 = 0.004$, strongly significant evidence against the null hypothesis.

Permutation testing was also applied using the *block statistic*

$$S(v_1) = v_1' B v_1 \quad (8.20)$$

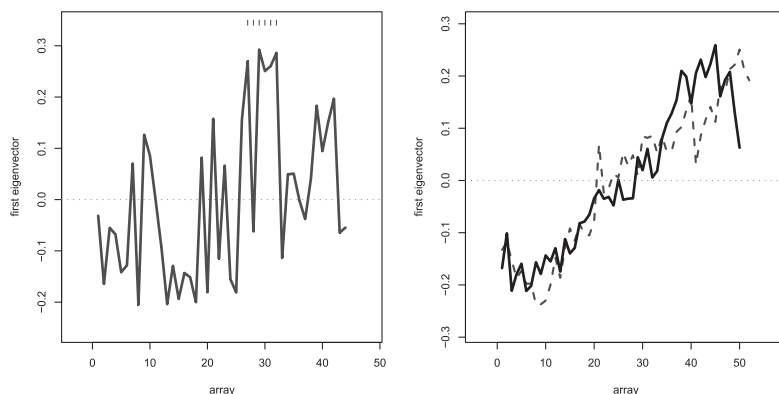


Figure 8.4 *Left panel:* Components of the first eigenvector plotted versus array index $1, 2, \dots, 44$, for the cardio data matrix; dashes emphasize the block of large components for arrays 27–32. *Right panel:* First eigenvector for control subject (solid line) and cancer patient (dashed line) matrices, prostate cancer study of Section 2.1; there was a drift in expression levels as the study progressed.

where B is the $n \times n$ matrix

$$B = \sum_h \beta_h \beta_h'. \quad (8.21)$$

The sum in (8.21) is over all vectors β_h of the form

$$\beta_h = (0, 0, \dots, 0, 1, 1, \dots, 1, 0, 0, \dots, 0)' \quad (8.22)$$

with the 1's forming blocks of length between 2 and 10 inclusive. $B = 5000$ permutations of the cardio vector v_1 gave only three v^{*b} values exceeding $S(v_1)$, p -value 0.0006, still stronger evidence against the i.i.d. null hypothesis.

The right panel of Figure 8.4 concerns the prostate data of Section 2.1. The first eigenvectors v_1 of $\hat{\Delta}$, computed separately for the 50 control subjects and the 52 cancer patients (with the two matrices individually doubly standardized), are plotted versus array number. Both vectors increase nearly linearly from left to right, indicating a systematic drift in expression levels as the study progressed. Some genes drifted up, others down, the average drift equaling zero because of standardization. The resulting correlation effect is small (or else $\hat{\alpha}$ would be larger than Table 8.3 indicates) but

genuine: both the slope and block tests decisively rejected independence across the columns.

Exercise 8.6 Show that if X is doubly standardized, the average linear regression coefficient $\hat{\beta}_i$ of x_{ij} on $j = 1, 2, \dots, n$ equals 0.

Table 8.4 *Correlation matrix and first eigenvector v_1 for the eight BRCA2 cases, $N = 3226$ genes. The last four arrays are strongly positively correlated.*

	1	2	3	4	5	6	7	8
1	1.00	.07	.09	.26	-.14	-.16	-.19	-.12
2	.07	1.00	.17	-.02	-.08	-.10	-.04	-.10
3	.09	.17	1.00	-.08	.01	-.06	-.10	-.03
4	.26	-.02	-.08	1.00	-.09	-.06	-.08	-.10
5	-.14	-.08	.01	-.09	1.00	.24	.34	.43
6	-.16	-.10	-.06	-.06	.24	1.00	.35	.35
7	-.19	-.04	-.10	-.08	.34	.35	1.00	.31
8	-.12	-.10	-.03	-.10	.43	.35	.31	1.00
v_1	.30	.15	.10	.19	-.45	-.44	-.46	-.48

Sometimes a pattern of column-wise correlation is obvious to the eye. Table 8.4 concerns the *BRCA data*, a microarray experiment concerning BRCA mutations, two known genetic factors that increase the risk of breast cancer. In this study, tumors from seven breast cancer patients having the BRCA1 mutation were compared against eight with BRCA2, on microarrays measuring $N = 3226$ genes. The table shows the 8×8 correlation matrix for the BRCA2 subjects, column standardized. We see strong positive correlations in the second group of four arrays, perhaps indicating an uncontrolled change in the development process. The first-four/last-four effect is particularly striking in the first eigenvector v_1 . A block test (now using all blocks of length 2 through 4) rejected independence at p -value 0.025, which is close to the most extreme result possible with only eight columns.

Two objections can be raised to our permutation tests: (1) they are really testing i.i.d., not independence; (2) non-independence might not manifest itself in the order of v_1 (particularly if the order of the microarrays has been shuffled in some unknown way).

Column-wise standardization makes the column distributions more sim-

ilar, mitigating objection (1). Going further, “quantile standardization” — say replacing each column’s entries by normal scores — makes the marginals exactly the same. The cardio data was reanalyzed using normal scores, with almost identical results.

Objection (2) is more worrisome from the point of view of statistical power. The order in which the arrays were obtained *should* be available to the statistician, and should be analyzed to expose possible trends like those in Figure 8.4. It would be desirable, nevertheless, to have independence tests that do not depend on order, that is, test statistics invariant under column-wise permutations.

This turns out to be problematic, for reasons having to do with Theorem 8.1. An obvious test statistic for column-wise independence, and one that doesn’t depend on the order of the columns, is

$$S = \sum_{j < j'} \hat{\Delta}_{jj'}^2 \binom{n}{2} \quad (8.23)$$

the average squared off-diagonal column correlation. But if X is doubly standardized then S is a monotone increasing function of $\hat{\alpha}$ (8.18), the estimated rms row correlation: so large values of S can always be blamed on large row-wise correlations.

Exercise 8.7 Express S as a function of $\hat{\alpha}$.

In other words, the spread of the solid histogram in Figure 8.1 can always be attributed to the spread of the line histogram. This doesn’t rule out all histogram tests: the histogram of the 946 off-diagonal values $\hat{\Delta}_{jj'}$ appears to be long-tailed to the right. Applying `locfdr` yielded five values $\hat{\Delta}_{jj'}$ having $\widehat{\text{fdr}}$ (6.38) less than 0.20, those exceeding 0.743. (Null density estimates $(\hat{\delta}_0, \hat{\sigma}_0, \hat{\pi}_0) = (-0.04, 0.236, 998)$ from the MLE method.) All five were from the block 27–32 emphasized in Figure 8.4.

The threshold 0.743 for significance seems remarkably high given that each $\hat{\Delta}_{jj'}$ is based on $N = 20\,426$ pairs $(x_{ij}, x_{ij'})$, as seen in Figure 8.3. However, the pairs are not independent of each other because of gene-wise correlations. We will see in Section 8.4 that the “effective sample size” for estimating $\Delta_{jj'}$ is *much* smaller than N .

8.4 Multivariate Normal Calculations

Classic statistical theory depended heavily on normal distributions. Normality is a risky assumption in large-scale studies but, risky or not, it is a

useful tool for analyzing our methodology. This section gives a brief review of the ideas involved.

A random $N \times n$ matrix X is said to have *matrix normal distribution*

$$X \sim \mathcal{N}_{N \times n} \left(M, \Sigma \otimes \Delta \right) \quad (8.24)$$

if the Nn entries x_{ij} are jointly normal, with means and covariances

$$E\{x_{ij}\} = m_{ij} \quad \text{and} \quad \text{cov}(x_{ij}, x_{i'j'}) = \sigma_{ii'} \Delta_{jj'}. \quad (8.25)$$

The notation $\otimes$ indicates a *Kronecker product*² as defined by (8.25). If $\Delta = I_n$, the $n \times n$ identity matrix, then

$$\text{cov}(x_{ij}, x_{i'j'}) = \begin{cases} \sigma_{ii'} & \text{if } j = j' \\ 0 & \text{if } j \neq j'; \end{cases} \quad (8.26)$$

that is, the columns of X are independent of each other, each with covariance matrix Σ , which was the assumption leading to our estimator $\hat{\alpha}$ in Section 8.2. If Δ is not diagonal, X has both its rows and columns correlated, a possibility we want to explore further.

Linear transformations are particularly convenient for matrix normal distributions. If

$$Y = A X B' \quad (8.27)$$

then

$$Y \sim \mathcal{N}_{a \times b} (AMB', A\Sigma A' \otimes B\Delta B'). \quad (8.28)$$

(The mean and covariance formulas in (8.28) depend only on (8.25) and do not require normality.)

Demeaning is an example of (8.27). Let $B = I_n - 1_n 1'_n/n$ and $A = I_N - 1_N 1'_N/N$, where 1_n is a column vector of n 1's. Then Y is the demeaned version of X , i.e., X with its row and column means subtracted off.

Exercise 8.8 (a) Verify this statement. (b) Show that Y has mean $M^Y = 0$ and covariance $\Sigma^Y \otimes \Delta^Y$, where

$$\sigma_{ii'}^Y = \sigma_{ii'} - \sigma_{i.} - \sigma_{.i'} + \sigma_{..} \quad \text{and} \quad \Delta_{jj'}^Y = \Delta_{jj'} - \Delta_{j.} - \Delta_{.j'} + \Delta_{..} \quad (8.29)$$

where the dots indicate averaging over the missing indices. (c) Show that demeaning reduces the rms correlations of both rows and columns.

² Many references reverse the notation to $\Delta \otimes \Sigma$, this being the convention for the R function `kronecker`, but both notations agree on (8.25).

In what follows we will assume that X has been both demeaned and doubly standardized, in the population sense that $M = 0$ and Σ and Δ have all diagonal elements 1,

$$m_{ij} = 0, \quad \sigma_{ii} = 1, \quad \text{and} \quad \Delta_{jj} = 1. \quad (8.30)$$

An unbiased estimator of α^2

The matrix normal model leads directly to an unbiased estimator of the squared rms correlation α^2 (8.15), *assuming that the columns of X are independent*, which, under (8.30), is equivalent to the column covariance matrix equaling the identity

$$\Delta = I_n. \quad (8.31)$$

As in (8.16), let $\bar{\alpha}$ be the observed rms row-wise correlation.

Theorem 8.3 *Under the matrix normal model (8.24), (8.30), (8.31),*

$$\tilde{\alpha}_n^2 = \frac{n}{n+1} \left(\bar{\alpha}^2 - \frac{1}{n} \right) \quad (8.32)$$

is an unbiased estimator of α^2 .

The proof appears at the end of this section. It depends on taking $\hat{\Sigma}$ (8.5) to have a Wishart distribution with n degrees of freedom. Under the usual normal assumptions, demeaning reduces the degrees of freedom to $n - 1$, suggesting the estimator

$$\tilde{\alpha}_{n-1}^2 = \frac{n-1}{n} \left(\bar{\alpha}^2 - \frac{1}{n-1} \right). \quad (8.33)$$

A comparison with our preferred estimator $\hat{\alpha}$ in (8.18) shows that they are proportional, with

$$\hat{\alpha}/\tilde{\alpha}_{n-1} = n/(n-1) \doteq 1 + 1/n. \quad (8.34)$$

Usually this small factor is negligible. Jensen's inequality says that $\tilde{\alpha}_{n-1}$ will be biased downward as an estimator of α , and simulations indicate that the slight increase in $\hat{\alpha}$ helps mitigate this bias.

Effective sample size

The $N = 20\,426$ points in Figure 8.3 appear strongly correlated, giving $\hat{\Delta}_{31,32} = 0.805$. There would be no questioning correlation if the points (x_{i32}, x_{i33}) were independent across cases $i = 1, 2, \dots, N$. But they are not independent, and their dependence dramatically lowers the effective sample size.

Assume again the matrix normal model (8.24), (8.30): $X \sim \mathcal{N}_{N \times n}(0, \Sigma \otimes \Delta)$ with the diagonal elements of both Σ and Δ equaling 1. Traditional multivariate analysis³ deals with the situation $\Sigma = I_N$, i.e., with the rows of X independent, say

$$\mathbf{x}_i \stackrel{\text{iid}}{\sim} \mathcal{N}_n(0, \Delta), \quad i = 1, 2, \dots, N. \quad (8.35)$$

This gives $\hat{\Delta}$ a scaled Wishart distribution, unbiased for Δ , whose n^2 means and $n^2 \times n^2$ covariances we can denote as

$$\hat{\Delta}_{n \times n} \sim \left(\Delta, \Delta^{(2)} / N \right) \quad \left[\Delta_{jk, lh}^{(2)} = \Delta_{jl} \Delta_{kh} + \Delta_{jh} \Delta_{kl} \right]. \quad (8.36)$$

So, for example, $\text{cov}(\hat{\Delta}_{11}, \hat{\Delta}_{23}) = 2\Delta_{12}\Delta_{13}/N$.

Correlation between the rows of X doesn't change the fact that $\hat{\Delta}$ is unbiased for Δ , but it increases the variability of $\hat{\Delta}$.

Theorem 8.4 Under the matrix normal model (8.24), (8.30), the column covariance estimator $\hat{\Delta} = X'X/N$ has mean and covariance

$$\hat{\Delta} \sim \left(\Delta, \Delta^{(2)} / N_{\text{eff}} \right) \quad (8.37)$$

where N_{eff} is the “effective sample size”

$$N_{\text{eff}} = N / \left[1 + (N-1)\alpha^2 \right] \quad (8.38)$$

where α is the rms correlation (8.15) and $\Delta^{(2)}$ is as in (8.36).

In other words, $\hat{\Delta}$ has the accuracy obtained from N_{eff} independent draws (8.35). For $N = 20\,426$ and $\alpha = \hat{\alpha} = 0.245$, the cardio estimate, we get $N_{\text{eff}} = 16.7$ (!). One has to imagine Figure 8.3 with only 17 points instead of 20 426 for a realistic impression of its accuracy.

Exercise 8.9 Show that the estimated standard error of $\hat{\Delta}_{31,32} = 0.805$ is at least 0.24, assuming $\alpha = 0.245$.

There is a tricky point to consider here: $\hat{\alpha}$ (8.18) is *not* a dependable estimate of α (8.15) if the columns of X are highly correlated. Theorem 8.1 works both ways: substantial column-wise correlations can induce an unwarranted impression of row-wise correlation. For the cardio study, in which column-wise correlation is strongly suspected, α may be considerably smaller than $\hat{\alpha}$.

It should be noted that formula (8.38) applies specifically to the estimation of Δ and that other estimation problems would have different effective

³ Most texts transpose the set-up so that the data matrix is $n \times N$, usually denoted $p \times n$, with the *columns* i.i.d.

sample sizes, perhaps not so conveniently depending on only the rms correlation α . The proof of Theorem 8.4 appears at the end of this section.

Correlation of t -values

Theorem 8.3 showed that if the columns of X were i.i.d. normal vectors, say

$$\mathbf{x}_j \stackrel{\text{iid}}{\sim} \mathcal{N}_N(\boldsymbol{\mu}, \Sigma) \quad j = 1, 2, \dots, n, \quad (8.39)$$

we could construct an unbiased estimator for Σ 's mean squared correlation α^2 . Our real interest, however, lies not in the individual columns but with summary statistics across the columns, like t_i and z_i in Section 2.1. Fortunately, the covariance matrix of summary statistics is often itself nearly Σ , so that $\tilde{\alpha}$ or $\hat{\alpha}$ remain relevant to them. Here we will give a specific result for t_i , the one-sample t -statistic.

For row i of X in (8.39) define

$$D_i = \sqrt{n}\bar{x}_i, \quad \delta_i = \sqrt{n}\mu_i, \quad \text{and} \quad S_i = \sum_{j=1}^n (x_{ij} - \bar{x}_i)^2 / (n-1), \quad (8.40)$$

$\bar{x}_i = \sum_j x_{ij}/n$, so that the t -statistic is

$$t_i = D_i / S_i^{1/2}. \quad (8.41)$$

Theorem 8.5 *The covariance between t_i and $t_{i'}$ under model (8.39) is*

$$\begin{aligned} \text{cov}(t_i, t_{i'}) &= \frac{n+1}{n-1} \begin{pmatrix} 1 & c_0 \rho_{ii'} \\ c_0 \rho_{ii'} & 1 \end{pmatrix} \\ &\quad + \frac{1}{2(n-1)} \begin{pmatrix} \delta_i^2 & \delta_i \delta_{i'} \rho_{ii'}^2 \\ \delta_i \delta_{i'} \rho_{ii'}^2 & \delta_{i'}^2 \end{pmatrix} + O\left(\frac{1}{n^2}\right) \end{aligned} \quad (8.42)$$

where

$$c_0 = 1 - (1 - \rho_{ii'}^2) / (2(n+1)). \quad (8.43)$$

In the null case, $\delta_i = \delta_{i'} = 0$,

$$\text{cor}(t_i, t_{i'}) = c_0 \rho_{ii'} + O(1/n^2). \quad (8.44)$$

(The proof is sketched at the end of this section.)

The effect of Theorem 8.4 is to make the t -statistic correlations just slightly smaller than those for the column entries. With $n = 20$ and $\rho_{ii'} = 0.4$, for example, we get $\text{cor}(t_i, t_{i'}) \doteq 0.392$. Numerical experimentation indicates the same kind of effect even in non-null cases, not surprisingly since the second term in (8.42) is of order only $O(1/n)$.

Proof of Theorem 8.3 With $\sigma_{ii'} \equiv 1$, we have $\rho_{ii'} = \sigma_{ii'}$ and $\bar{\alpha}^2$ (8.16) equaling

$$\bar{\alpha}^2 = \frac{\sum_i \sum_{i'} \hat{\sigma}_{ii'}^2 - N}{N(N-1)} \quad (8.45)$$

compared to

$$\alpha^2 = \frac{\sum_i \sum_{i'} \sigma_{ii'}^2 - N}{N(N-1)}. \quad (8.46)$$

A standard Wishart result, as in (8.36), says that $\hat{\sigma}_{ii'}^2$ has mean and variance

$$\hat{\sigma}_{ii'}^2 \sim \left(\sigma_{ii'}^2, \frac{1 + \sigma_{ii'}^2}{n} \right) \quad (8.47)$$

so that

$$E \left\{ \sum_i \sum_{i'} \hat{\sigma}_i^2 \right\} = \frac{n+1}{n} \sum_i \sum_{i'} \sigma_{ii'}^2 + \frac{N^2}{n}. \quad (8.48) \quad \square$$

Exercise 8.10 Complete the proof of Theorem 8.3.

Proof of Theorem 8.4 The covariance calculation for $\hat{\Delta} = X'X/N$ involves the expansion

$$\begin{aligned} \hat{\Delta}_{jk} \hat{\Delta}_{lh} &= \left(\sum_i X_{ij} X_{ik} / N \right) \left(\sum_{i'} X_{i'l} X_{i'h} / N \right) \\ &= \frac{1}{N^2} \left(\sum_i X_{ij} X_{ik} X_{il} X_{ih} + \sum_{i \neq i'} X_{ij} X_{ik} X_{i'l} X_{i'h} \right). \end{aligned} \quad (8.49)$$

Using the formula

$$E\{Z_1 Z_2 Z_3 Z_4\} = \gamma_{12} \gamma_{34} + \gamma_{13} \gamma_{24} + \gamma_{14} \gamma_{23} \quad (8.50)$$

for a normal vector $(Z_1, Z_2, Z_3, Z_4)'$ with means zero and covariances γ_{ij} , (8.25) gives

$$E \left\{ \sum_i X_{ij} X_{ik} X_{il} X_{ih} \right\} = N \left[\Delta_{jk} \Delta_{lh} + \Delta_{jl} \Delta_{kh} + \Delta_{jh} \Delta_{kl} \right] \quad (8.51)$$

(using $\sigma_{ii'} = 1$) and

$$\begin{aligned} E \left\{ \sum_{i \neq i'} X_{ij} X_{ik} X_{i'l} X_{i'h} \right\} &= N(N-1) \Delta_{jk} \Delta_{lh} \\ &\quad + (\Delta_{jl} \Delta_{kh} + \Delta_{jh} \Delta_{kl}) \sum_{i \neq i'} \sigma_{ii'}^2. \end{aligned} \quad (8.52)$$

Then (8.49) yields

$$E\{\hat{\Delta}_{jk}\hat{\Delta}_{lh}\} = \Delta_{jk}\Delta_{lh} + (\Delta_{jl}\Delta_{kh} + \Delta_{jh}\Delta_{kl})\left(\frac{1 + (N-1)\alpha^2}{N}\right) \quad (8.53)$$

giving

$$\text{cov}(\Delta_{jk}, \Delta_{lh}) = (\Delta_{jh}\Delta_{kl} + \Delta_{jl}\Delta_{kh})/N_{\text{eff}} \quad (8.54)$$

as in (8.37). $\square$

Proof of Theorem 8.5 Letting $D = (D_i, D_r)'$, $\delta = (\delta_i, \delta_r)'$, and $S = (S_i, S_r)'$, we have

$$D \sim \mathcal{N}_2\left(\delta, \begin{pmatrix} 1 & \rho_{ir} \\ \rho_{ir} & 1 \end{pmatrix}\right) \quad \text{and} \quad S \sim \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \frac{2}{n-1} \begin{pmatrix} 1 & \rho_{ir}^2 \\ \rho_{ir}^2 & 1 \end{pmatrix}\right) \quad (8.55)$$

with D and S independent. The notation indicates the mean and covariance of S , obtained from (8.36). Also defining $t = (t_i, t_r)'$ and $U = (S_i^{-1/2}, S_r^{-1/2})'$,

$$t = D \cdot U, \quad (8.56)$$

where the dot indicates component-wise multiplication. A standard two-term Taylor expansion gives approximate mean and variance for U ,

$$U \sim \left(\left(1 + \frac{3}{4(n-1)}\right)\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \frac{1}{2(n-1)} \begin{pmatrix} 1 & \rho_{ir}^2 \\ \rho_{ir}^2 & 1 \end{pmatrix}\right). \quad (8.57)$$

Because D and U are independent in (8.56), we can use the general identity

$$\text{cov}(t) = \text{cov}(D) \cdot E\{UU'\} + (E\{D\}E\{D\}') \cdot \text{cov}(U) \quad (8.58)$$

to finish the proof. $\square$

Exercise 8.11 (a) Derive (8.57). (b) Verify (8.58). (c) Complete the proof.

8.5 Count Correlations

Correlations among the components of $\mathbf{z}$, the vector of z -values, lead to correlations in the count vector $\mathbf{y}$ (7.4). This is more than a theoretical curiosity since it directly affects the interpretation of false discovery rates (Reason (III), Section 6.4).

Here we will trace the effects of correlation on counts of the null cases, those z -values having $z_i \sim \mathcal{N}(0, 1)$. Figure 8.5 used program `simz`, Section 8.2, to generate 2000 simulated $\mathbf{z}$ vectors, each of length $N = 4000$ and rms correlation $\alpha = 0.20$, all N cases null. The figure plots Y_1 versus Y_0 , where

$$Y_0 = \#\{|z_i| \leq 1\} \quad \text{and} \quad Y_1 = \#\{z_1 \geq 2\}. \quad (8.59)$$

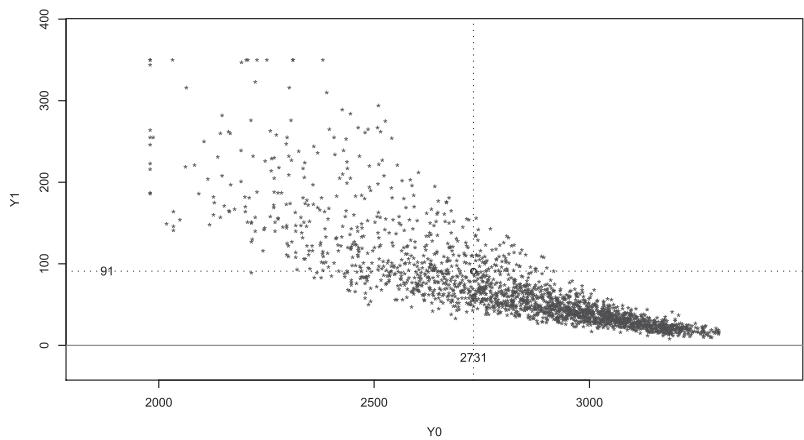


Figure 8.5 Tail counts $Y_1 = \#\{z_i \geq 2\}$ plotted versus central counts $Y_0 = \#\{|z_i| \leq 1\}$ for 2000 simulated z vectors with $N = 4000$, $\alpha = 0.20$. Large values of Y_0 give small values of Y_1 , while small Y_0 gives large Y_1 .

Table 8.5 *Standard deviations and correlation for Y_0 and Y_1 (8.59) for the situation in Figure 8.5 ($\alpha = 0.20$) and also for independent z -values ($\alpha = 0$).*

	$\alpha = .20$		$\alpha = 0$	
	Y_0	Y_1	Y_0	Y_1
standard deviation	272.0	57.3	30.1	8.89
correlation	-.82		-.26	

There is a very strong negative correlation: if the number of central counts Y_0 much exceeds its expectation 2731, then the number of tail counts Y_1 is less than its expectation 91, and conversely. Table 8.5 shows that the negative correlation is far stronger than in the situation where the z -values are independent.

In an actual large-scale analysis, we might be interested in the false discovery rate for region $\mathcal{Z} = [2, \infty)$, for which Y_1 would be the unobserved numerator $N_0(\mathcal{Z})$ in the false discovery proportion (2.28). The fact that central counts help predict tail counts motivates use of the empirical null

distributions, Section 6.2. What goes wrong if we ignore this effect is illustrated in Figure 6.4.

The striking correlation seen in Figure 8.5 is predicted by the rms approximation (7.38). Combined with Lemma 7.1, this suggests that the null count vector $\mathbf{y}$ is distributed approximately as

$$\mathbf{y} = \mathbf{y}_0 + Nd\mathbf{W}A. \quad (8.60)$$

Here $\mathbf{y}_0$ is a multinomial vector with expectation $\boldsymbol{\nu}_0 = Nd(\dots, \varphi(x_k), \dots)'$ and covariance $\text{diag}(\boldsymbol{\nu}_0) - \boldsymbol{\nu}_0\boldsymbol{\nu}_0'/N$, this being cov_0 in (7.14); $\mathbf{W}$ is the “wing-shaped function”

$$W_k = \frac{\varphi''(x_k)}{\sqrt{2}} = \varphi(x_k) \frac{x_k^2 - 1}{\sqrt{2}} \quad \text{for } k = 1, 2, \dots, K, \quad (8.61)$$

illustrated in Figure 8.6; and A is a random variable, independent of $\mathbf{y}_0$, with mean and variance

$$A \sim (0, \alpha^2). \quad (8.62)$$

Exercise 8.12 Show that $\mathbf{y}$ in (8.60) has the mean and covariance suggested by Lemma 7.1 and (7.38).

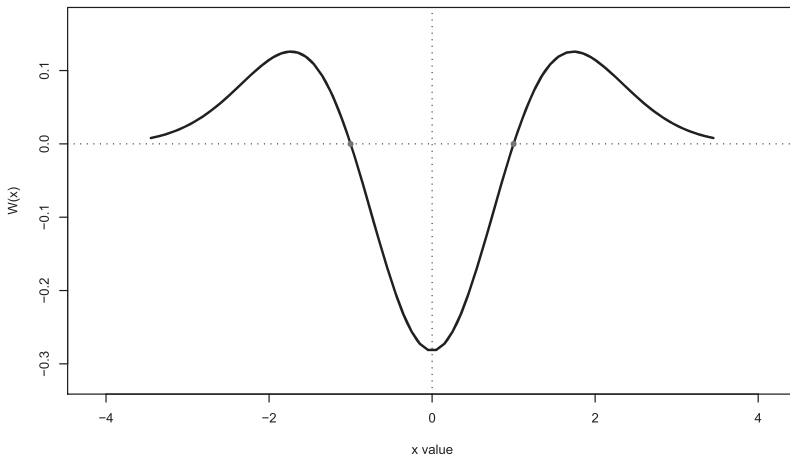


Figure 8.6 Wing-shaped function $\varphi(x)(x^2 - 1)/\sqrt{2}$.

A positive value of A in (8.60) is seen to depress the count vector $\mathbf{y}$ for x in the interval $[-1, 1]$ and to increase it outside that interval: in other words, it decreases the central count Y_0 and increases tail counts such as

Y_1 . Negative A has the opposite effects. This accounts nicely for Figure 8.5. The magnitude of the negative correlation effects depends on α in (8.62), which controls the size of A 's excursions from 0.

Notes

Large-scale correlation problems have attracted considerable current interest, as indicated in the notes for Chapter 7. Most of this chapter's material comes from Efron (2009a), where tests for column-wise independence are investigated more thoroughly, and Efron (2007a). Muralidharan (2010) devised the nice unbiasedness result of Theorem 8.3. Olshen and Rajaratnam (2009) show, by advanced methods, that the double standardization argument converges, except for small, artificially constructed matrices X .

The BRCA data comes from Hedenfalk et al. (2001). Ashley et al. (2006), working out of the laboratory of T. Quertermous, analyze the cardio data (Agilent two-color hybridization arrays), while the snp data (unpublished) is also from the Quertermous team. Program `simz` is referenced in Appendix B.